The Political Economy of Greenwashing

Spatialising the Myth of Immateriality in Virtual Production and Digital Spatial Fix

Hanpu Li

Fig. 1. Promotional imagery for Epic Games' Virtual Production curriculum (Animation World Network, 2020).

Research Question & Contribution

Research Question

How does virtual production, framed as a low-carbon digital alternative, reorganise land use, energy demand and spatial governance at the metropolitan periphery?

Contribution

Conceptually: situates virtual production within the Digital Spatial Fix

Methodologically: develops a replicable forensic spatial audit pipeline

Empirically: quantifies spatial, discursive and economic asymmetries in the Shepperton expansion

Situation

Growth of Virtual Production in UK screen industries

Framed as sustainable alternative to location shooting

Case: Shepperton Studios Expansion

Virtual production is widely framed as a decarbonisation strategy.

Virtual production operates as a technological veneer for renewed rounds of spatial expansion and resource appropriation.

Theoretical Framework: a Marxist Political Economy Approach

Media Materiality (Bozak, 2012)

Digital media always depend on physical infrastructures and resource extraction.

The Jevons Paradox (Sharma, 2024)

Efficiency gains can increase total energy consumption.

The Digital Spatial Fix (Greene & Joseph, 2015)

Virtual production functions as a *capitalist spatial fix*: a strategy through which capital resolves its own crises by expanding into new territories, infrastructures and ecological capacities.

Accumulation by Dispossession (Harvey, 2004)

Digital expansion requires the continual appropriation of land, energy and ecological capacity.

Methodology: Audit Pipeline

> 01_RAW_MINING: Overpass API (OSM)

> 02_SPATIAL_PROJECTION: Python (WGS84 → EPSG:27700)

> 03_DISCOURSE_AUDIT: Retrieval-Augmented Critical Discourse Analysis (CDA)

[Corpus: Spelthorne Municipal Grey Literature & Planning Dossiers]

Terminal Output:

```
AGGREGATE_LOGISTICAL_SPRAWL: 132,123.12 SQM
TOTAL_LAND_CONVERSION: 13.2123 Hectares
TOTAL_POWER_ANCHORS: 17
```

```
(audit_env) caitlye@Caits-MacBook-Pro-2 Desktop % python "Spatial-Projection-Audit-Script.py"
--- INITIATING EMPIRICAL SPATIAL AUDIT (OSM_AUDIT_2025) ---
SHEPPERTON_SECTOR: 109,411.89 SQM | Nodes: 2
LONGCROSS_SECTOR:  22,711.22 SQM | Nodes: 15
--------------------------------------------------
AGGREGATE LOGISTICAL SPRAWL: 132,123.12 SQM
TOTAL LAND CONVERSION:      13.2123 Hectares
TOTAL POWER ANCHORS:        17
--- AUDIT COMPLETE: DATA READY FOR VISUALISATION ---
```

Semantic Deconstruction:

"The proposed development is in a highly sustainable location and will provide a range of social, economic and environmental benefits." (Turley, 2018)

"The prospective applicant has indicated an intention to improve the ecological condition… and to deliver improvements to the wider site in terms of biodiversity." (Spelthorne Borough Council, 2018)

132,123.11 SQM Logistical Sprawl (Asphalt expansion)
5.4 GWh/year (regulated demand, excluding tenant fit-out and unregulated loads)
79+ Mature Trees Removed (as per Environmental Statement, 2018)
£6,150 S106 Buyout (vs. £392,000,000 Annual GVA)

Fig. 2. Terminal output of the spatial audit pipeline (OSM_AUDIT_2025). Aggregate metrics calculate total land conversion at 13.2123 hectares. Author's own execution.

Methodological Note (Data Scope & Limits)

OSM extraction date: March 2026

Logistical sprawl defined as: parking, service roads, impermeable service yards

Power anchors identified via substation and high-capacity grid nodes tagged in OSM

NO2 projection based on AADT modelling from i-Transport (2018) baseline

OSM data accuracy ~85-95% for tagged infrastructure (per OSM wiki); projection error <0.1% at EPSG:27700 scale.

Problem

The problem lies in the hidden material expansion behind claims of immateriality.

Spatial-Ecological Imbalance

Aggregate Logistical Sprawl: 132,123.11 sqm

Parking Bays: 2845

Total Land Conversion: 13.2123 Hectares*

Traffic Rebound (AADT): +703 Vehicles/Day (Source: i-Transport, 2018, Table 5.14)

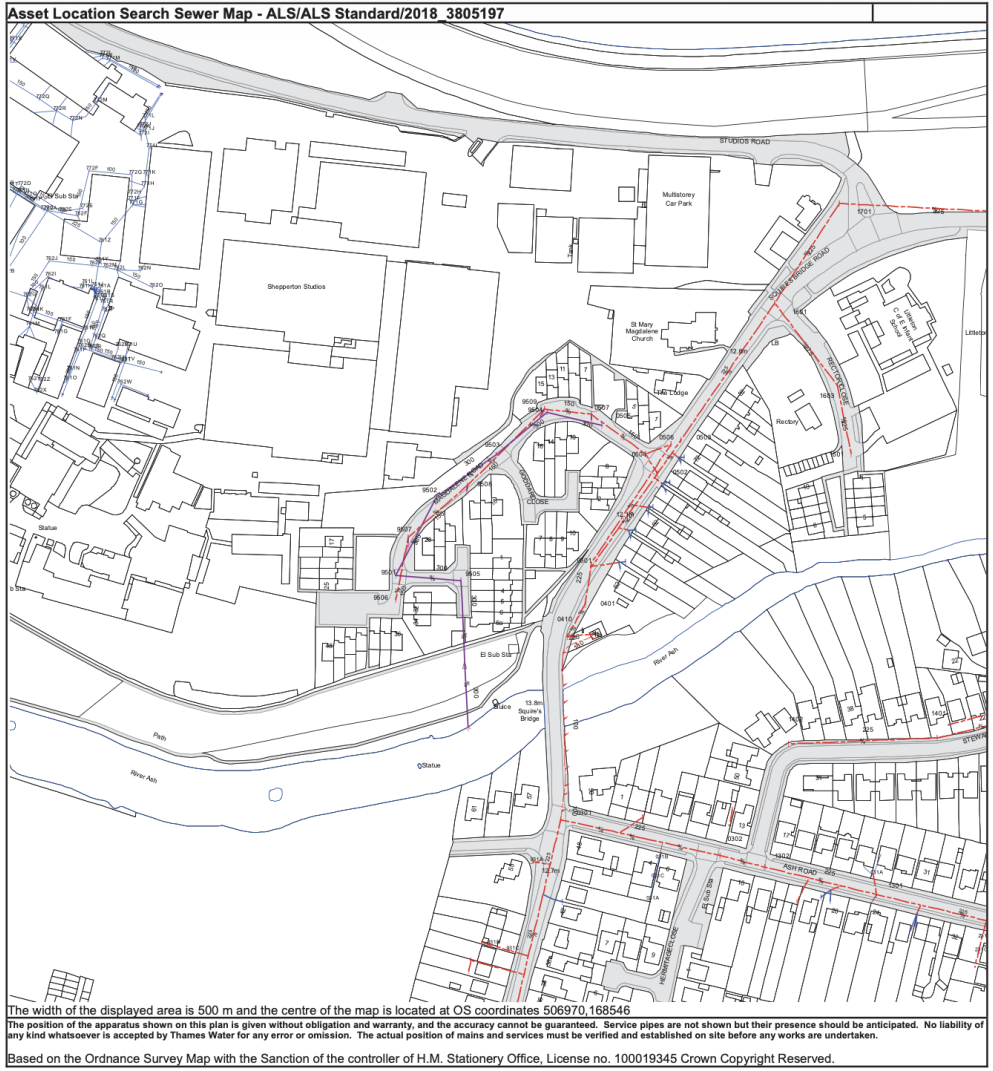
Atmospheric Compermise: NO2 +2.19 $\mu g/m^3$ (Source: Turley, ES Vol 1, Air Quality Assessment)

132 kV Nodes: 17

This logistical sprawl constitutes the spatial fix demanded by digital capital to sustain accumulation cycles.

*Equivalent to paving over 18.5 professional football pitches.

Geovisualisation of Shepperton expansion vectors. Data sourced from OSM_AUDIT_2025 repository.



For each mid-size production:

Base Power Draw: 2x 1200 Amp + 2x 250 Amp Diesel Generators

Combustion rate: 296 Litres of Diesel / Day

Climate Cost: 52 Tonnes GHG Emissions (CMPA / Felder et al., 2022)

Source: Asset Location Search Map (Hydrock, 2018, p. 6).

Implications (Preview)

Environmental: loss of biomass, NDVI collapse, heat accumulation

Economic: 5.4 GWh/year new loads, escalation of local infrastructure cost

Governance: discourse-led legitimisation of expansion

Land, Flood Risk & Ecological Loss

High-Capacity Grid Dependence: 5.4 GWh/Year Regulated Energy Demand

5,396,561 kWh

Hydrological Control : 39 Hectares in Flood Zone 2 Hardsurfaced

Physical Erasure: 79+ MATURE TREES REMOVED

This reflects what political ecology calls a *violent discipline of nature.*

Arboricultural Impacts: Summary (For details, see below)	
Impact	**No. of Trees**
Trees to be removed	79
Groups of trees to be removed	18
Groups of trees to be partially removed	4
TPO trees to be removed	0
Trees to be pruned	0
Trees where manual excavation may needed within RPAs	9
Trees where above soil surfacing may needed within RPAs	6
Trees with proposed underground services within RPAs	0

Table 1. Summary of arboricultural impacts. Extracted from Spelthorne Borough Council Planning Dossier 18/01212/OUT (Arboricultural Implications Report, August 2018).

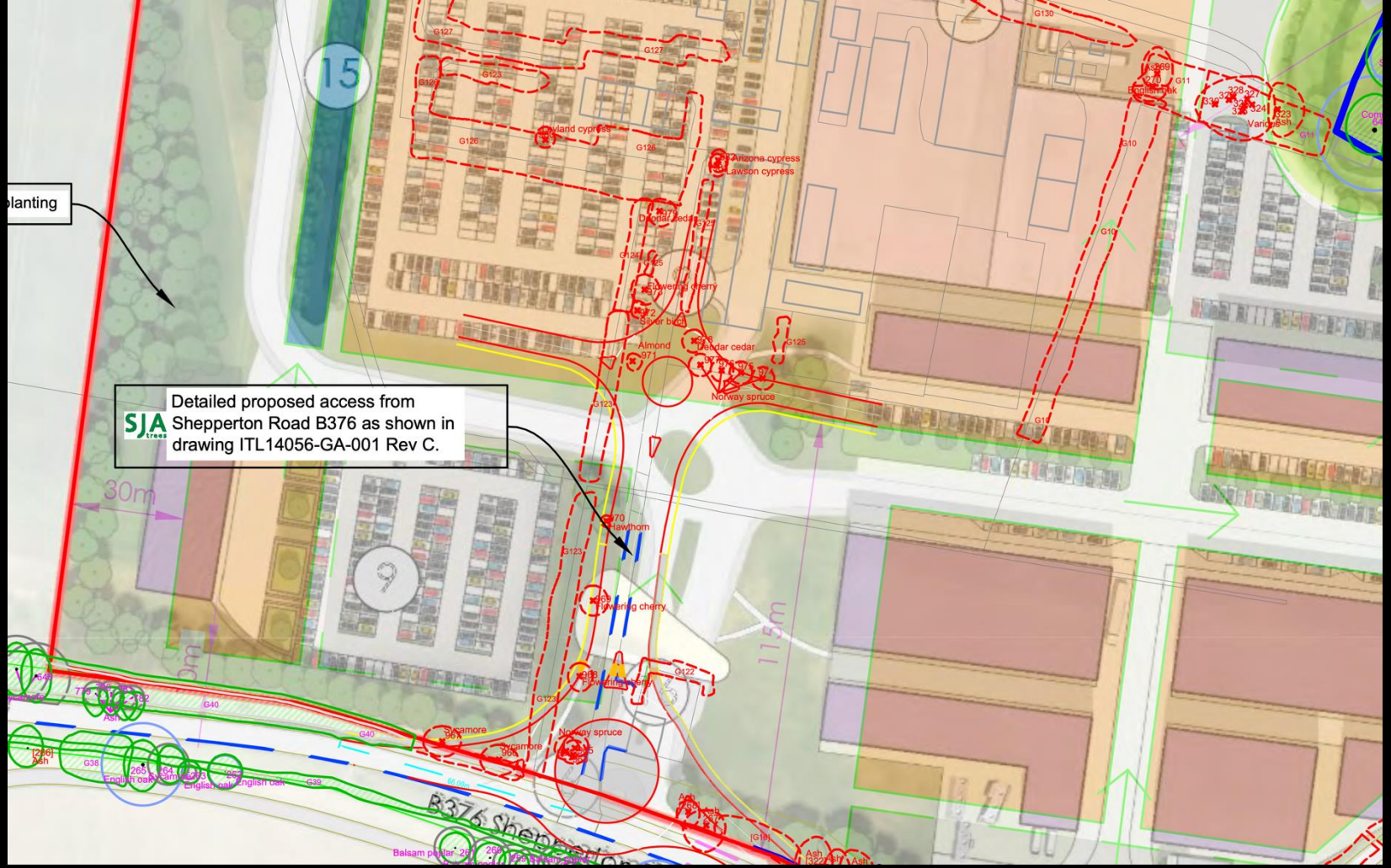


Fig 2: SJA Trees (2018) Tree Protection Plan. Drawing No: SJA TPP 18158-02.

Institutional Complicity & Moral Licensing

£392,000,000

Annual Net GVA (Capital Fix)

(Turley Economics, 2018, Table 6.2)

£6,150*

S106 Ecological Buyout Fee

(Spelthorne Borough Council, 2019)

(0.0015%)

***For 13.2 hectares, that means 4.6 pence per square metre.**

Feasibility

Uses already available spatial data (OSM)

The audit methodology leverages open-source Volunteered Geographic Information, specifically OpenStreetMap via the Overpass API, to extract hard spatial metrics. This circumvents the industry's 'selective visibility' by utilising publicly accessible geographic data to quantify physical infrastructure, such as high-voltage nodes and logistical sprawl, without relying on proprietary corporate disclosures.

Can be integrated into existing EIA processes

This forensic framework is highly feasible as it directly interfaces with the statutory frameworks already mandated by the EIA Regulations 2017. By applying Natural Language Processing and spatial quantification to standard planning dossiers, local authorities can systematically reverse-read institutional euphemisms to expose the exact material costs, such as energy demand and habitat erasure, already documented within developer submissions. (Spelthorne Borough Council, 2018; Turley, 2018)

Scalable across UK production hubs

Because the spatial projection utilises the standardised British National Grid (EPSG:27700), this algorithmic audit is universally scalable across the entire UK screen sector. It can be systematically deployed to quantify the cumulative spatial fix and metabolic rift across expanding megahubs, tracking continuous capital territorialisation from Shepperton down to the Longcross and Leavesden clusters. (PwC, 2018; Greene and Joseph, 2015)

Epistemological Limits & Future Trajectories

1. Geospatial: Time-Series Remote Sensing

Limit: OSM latency (Jan ‘26) and static mapping.
Why Not: Deploy Google Earth Engine & Sentinel 2 data for NDVI verification.

2. Thermodynamic: Micro-Attribution

Limit: Macro-grid aggregation.
Why Not: HVAC & server cooling load regression modelling.

3. Algorithmic: The "Hypocrisy Index"

Limit: Manual CDA bottleneck.
Why Not: Automated NLP scraping of corporate ESG reports.

4. Logistical: Telemetric Falsification

Limit: Reliance on 2018 EIA predictive models.
Why Not: Live Traffic API monitoring (TomTom/Google Maps Traffic).

So Why Not…

Part II:
Earth Observation & Metabolic Quantification

Empirical Quantification of Biophysical Erasure and Localised Thermal Anomalies

(Bonus)

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ST & EO Pipeline (GEE API)

1. Core EO Datasets
ESA Sentinel-2 (S2_SR_HARMONIZED): 10 m multispectral imagery for vegetation dynamics
USGS Landsat 8 (Collection 2, Tier 1 - TIRS): 100 m thermal infrared data for land-surface temperature

2. Cloud-Masking & Temporal Normalisation
-**GEE QA60 bitwise cloud-masking** to remove atmospheric interference.
-**365-day rolling mean smoothing** (Python: `pandas`, `NumPy`) to eliminate seasonal meteorological noise and expose structural ecological trends.
-`dropna()` **filtering** to purge cloud-induced null artefacts in thermal data.

3. Minimal Open-Source Processing Stack
Google Earth Engine JavaScript API for NDVI and LST computation.
Python for local regression, time-series cleaning, and chart rendering.
Entire workflow is **fully open-source**, scalable, and reproducible without proprietary GIS software.

NDVI Collapse & Thermodynamic Dissipation

extractcsv.py | NDVI.js

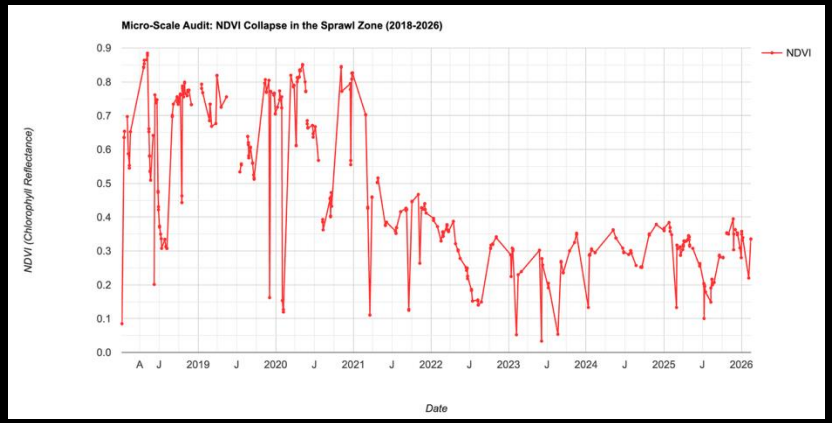
Users > caitlye > Desktop > geographic > raw > NDVI.js > ndviCollection > sentinel2.map() callback

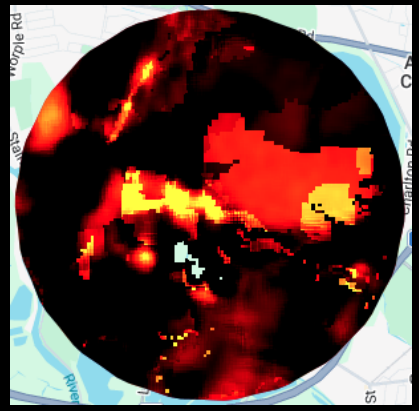
```javascript
// Shepperton NDVI
// 1. 宏观缓冲带 (1km)
var shepperton = ee.Geometry.Point([-0.4640, 51.4065]);
var analysisBuffer = shepperton.buffer(1000);

// 2. 微观灾区
var sprawlZone = ee.Geometry.Point([-0.469366, 51.410315]).buffer(100);

var senti any : ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
  .filterBounds(analysisBuffer)
  .filterDate('2018-01-01', '2026-03-04')
  .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20));

function maskS2clouds(image) {
  var qa = image.select('QA60');
  var cloudBitMask = 1 << 10;
  var cirrusBitMask = 1 << 11;
  var mask = qa.bitwiseAnd(cloudBitMask).eq(0).and(qa.bitwiseAnd(cirrusBitMask).eq(0));
  return image.updateMask(mask).divide(10000);
}

// 直接在原图上计算 NDVI
var ndviCollection = sentinel2.map(function(img) {
  var masked = maskS2clouds(img);
  var ndvi = masked.normalizedDifference(['B8', 'B4']).rename('NDVI');
  return masked.addBands(ndvi).copyProperties(img, ['system:time_start', 'system:index']);
}).select('NDVI');

// 只统计 Sprawl Zone 的 NDVI 均值
var tsChart = ui.Chart.image.series(ndviCollection, sprawlZone, ee.Reducer.mean(), 10)
  .setOptions({
    title: 'Micro-Scale Audit: NDVI Collapse in the Sprawl Zone (2018-2026)',
    vAxis: {title: 'NDVI (Chlorophyll Reflectance)', min: 0, max: 0.8},
    hAxis: {title: 'Date'},
    lineWidth: 2,
    pointSize: 3,
    colors: ['#FF3333'] // 把折线变成警示的血红色
  });
print(tsChart);

// 损失图 (2018基线 vs 2026现状)
var baseline2018 = ndviCollection.filterDate('2018-01-01', '2019-01-01').mean();
var recent2026 = ndviCollection.filterDate('2025-09-01', '2026-03-04').mean();
var loss = baseline2018.subtract(recent2026).rename('NDVI_Loss');

Map.centerObject(shepperton, 15);
```

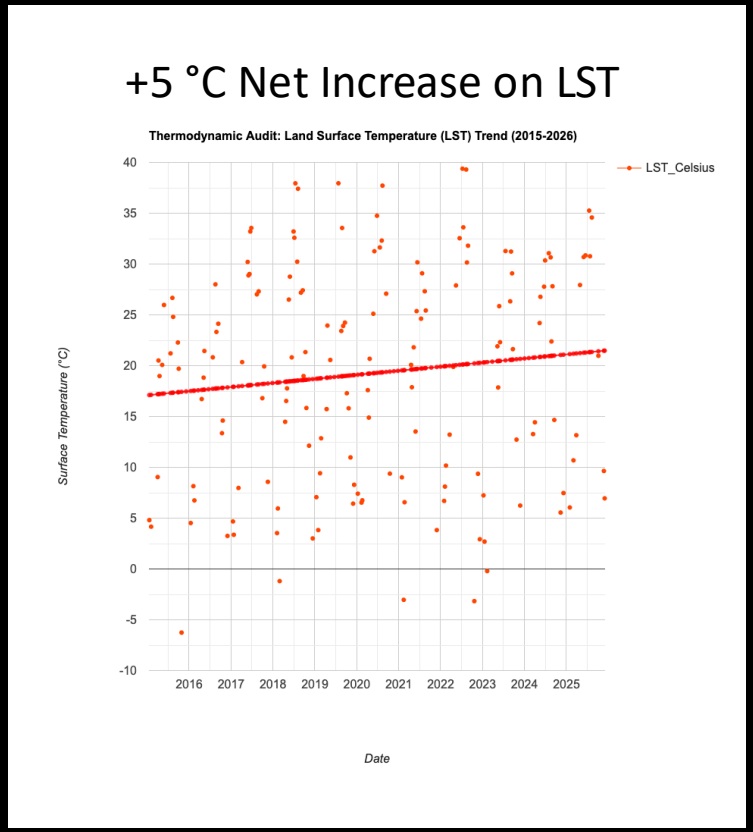
Ln 23, Col 51 | Spaces: 2 | UTF-8 | LF | JavaScript | Continue (NE)

Source Code: Li, H. (2026) *Shepperton NDVI Spatiotemporal Audit Algorithm*. Accessible at: https://code.earthengine.google.com/0cd023633ba069e4320a7a081bd65b62

NDVI Collapse & Thermodynamic Dissipation







Source: Author's calculation using Copernicus Sentinel-2 (Harmonised) data (ESA, 2026) via Google Earth Engine (Gorelick *et al.*, 2017).

Source: Author's calculation using Copernicus Sentinel-2 Harmonized data (ESA, 2026) via Google Earth Engine (Gorelick *et al.*, 2017).

Transparency, Reproducibility and Replicability

Repository: github.com/HanpuLi/-OSM_AUDIT_2025

Data Extraction Date: 3 March 2026

Spatial Projection: British National Grid (EPSG:27700)

Pipeline Built Via: Overpass API (OSM Query)
Python (*geopandas*, *matplotlib*)
Google Earth Engine API:
ESA Sentinel-2 (S2_SR_HARMONIZED) and
USGS Landsat 8 (Collection 2, Tier 1 – TIRS)

Conclusion: Policy & Structural Solutions

1. Integrate Spatial Audits into Planning Approval Processes

Implementation:
>Require forensic spatial audits as mandatory appendix to EIA submissions
>Planning authorities adopt standardised OSM-based audit protocols
>Embed audit results in publicly accessible planning portals

Rationale:
>Current EIA projections (e.g., Shepperton 2018) rely on developer-commissioned models without independent spatial verification
>Standardised audits enable cross-site comparison and prevent greenwashing claims

Conclusion: Policy & Structural Solutions

2. Mandate Lifecycle Spatial Accounting for Virtual Production

Implementation:
>Production companies report annual land footprint and energy consumption
>Data aggregated in national Virtual Production Environmental Registry
>Five-year audits compare actual vs. projected environmental impact

3. Link Tax Incentives to Measurable Carbon and Land-Use Caps

Implementation:
>UK Film Tax Relief eligibility conditional on:
- Maximum 2 hectares new hard-surfacing per production hub
- 100% renewable energy procurement for virtual stages
>Tiered tax rates: 25% relief for compliant studios, 15% for non-compliant

Rationale:
>Current £392M GVA benefits from tax incentives without environmental conditionality
>Caps incentivise technology efficiency rather than spatial expansion

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Thank You.

Scan to access
OSM_AUDIT_2025
GitHub Repository
EPSG:27700
Logic & Raw JSON Datasets

github.com/HanpuLi/-OSM_AUDIT_2025

Evaluation: Counter-Arguments

1. Doesn't Virtual Production fundamentally save carbon by drastically reducing global aviation emissions from location shooting?

2. You cite 13.2 hectares of asphalt and 17 power nodes. How can you scientifically attribute this massive physical sprawl specifically to 'Virtual Production'?

3. Your audit is entirely critical. But the EIA projects £392 Million in
Isn't this economic injection worth the sacrifice of 79 trees?

Doesn't Virtual Production fundamentally save carbon by drastically reducing global aviation emissions from location shooting?

I: The Aviation Fallacy

Efficiency in computation triggers the Rebound Effect (Sharma, 2024). Savings from avoided flights are reinvested into higher-resolution rendering and local physical sprawl. Aviation emissions are simply displaced into 5.4 GWh of local grid parasitism.

You cite 13.2 hectares of asphalt and 17 power nodes. How can you scientifically attribute this massive physical sprawl specifically to 'Virtual Production'?

II: The Attribution Trap

Virtual Production operates as a "Digital Spatial Fix" (Greene & Joseph, 2015). You cannot separate the digital pixel from the physical brick.

This audit does not claim that 100% of land conversion is causally reducible to virtual production. Rather, it examines how virtual production discourse legitimises and stabilises expansionary spatial restructuring.
A full counterfactual baseline would require longitudinal comparative studio data, which exceeds the scope of this project.

Your audit is entirely critical. But the EIA projects £392 Million in Net GVA.

Isn't this economic injection worth the sacrifice of 79 trees?

III: The Economic Imperative

The audit does not deny the wealth generation; it exposes the structural asymmetry of its extraction (Harvey, 2004). Economic gains are privatised, while ecological burdens are socialised, a classic asymmetry within capitalist accumulation.

Under strict land-use and energy caps, virtual production could contribute to decarbonisation; the issue lies in its current governance framework.

Evaluation Summary

Q1. Does VP save carbon via reduced aviation?
A. Savings rebound into compute and local sprawl (5.4 GWh).

Q2. Can sprawl be attributed to VP?
A. VP functions as a Digital Spatial Fix; discourse legitimises expansion

Q3. Isn't £392m GVA worth 79 trees?
A. Value extraction is asymmetric; caps and governance matter, not growth alone.

Appendix

Here are unused slides.

Institutional Complicity & Moral Licensing

£1,096,500,000

(SBC Municipal Debt)

£100k + £500k

Business Hub Sponsorship + Community Fund

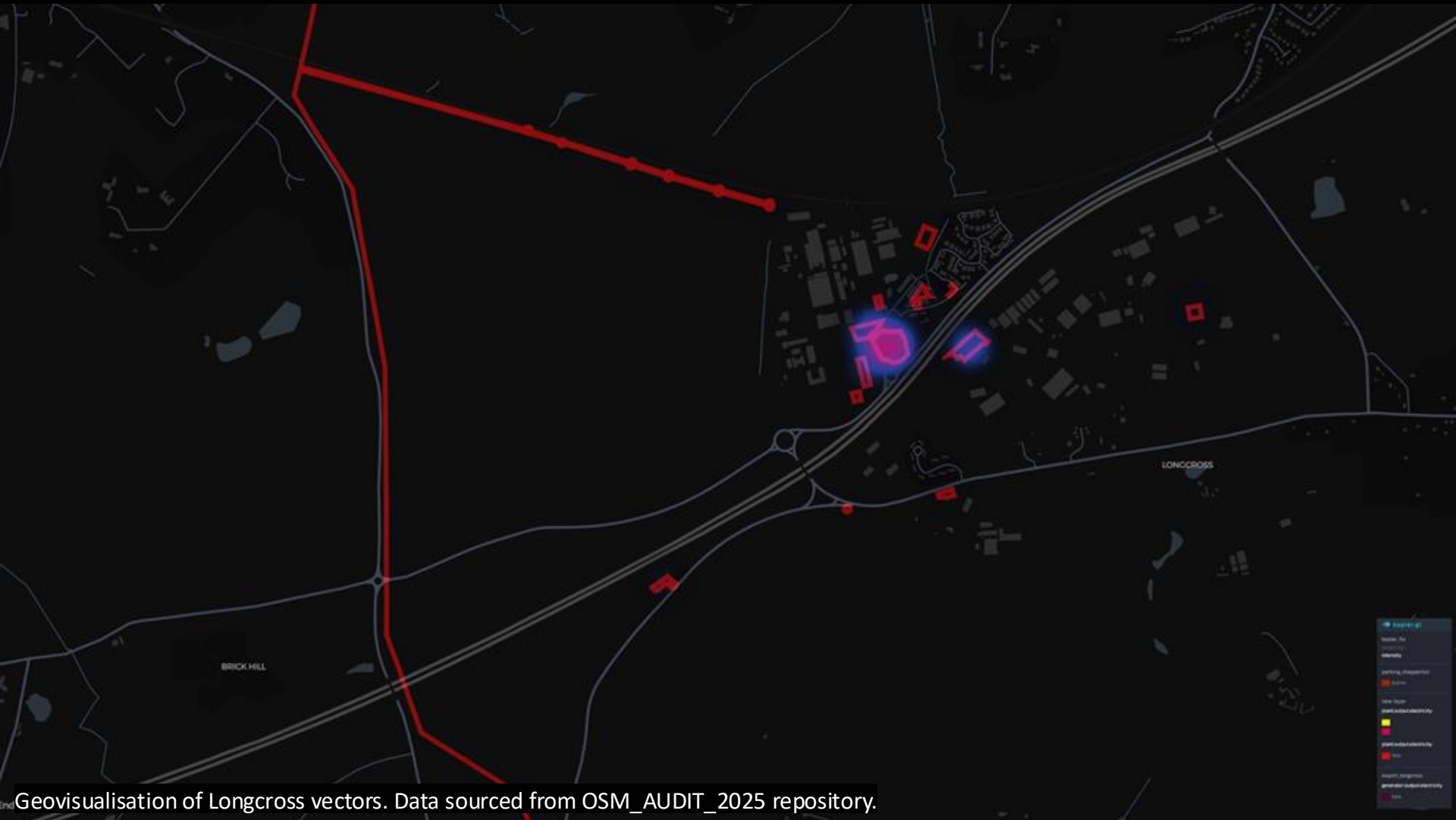
Data Sources: CIPFA (2023, p.1); Spelthorne Borough Council (2019; 2025).

“Very Special Circumstances”

Overriding greenbelt protection via economic justification

Application Dates	Valid: 10.04.2025	Expiry: 05.06.2025	Target: Extension of time agreed to 30.06.2025
Executive Summary	The proposal is for the use of the site for temporary car parking over the period mid-May to mid-November. The site is located within the Green Belt and the applicant has not presented material considerations that the Planning Authority (LPA) would consider to represent ‘Very Special Circumstances’. However, recent events in the film industry and geopolitically, together with the Government’s support for the UK economy and film making, in particular, are considered to provide very special circumstances that can be weighed against the harm. Overall, in this particular situation, at this specific time, it is considered that the potential harm to the Green Belt by reason of inappropriateness, and any other harm resulting from the proposal, is clearly outweighed. The design and appearance of the temporary use reflects the existing use of the site and cumulatively for a temporary period is not considered to represent a significant adverse impact over the existing situation. The likely noise created by the proposal for a temporary period is not considered to have a significant material impact on the adjoining properties. The car parking is displaced parking, not additional parking and therefore the proposal is not considered to represent a significant material impact to highway safety or the overall highway network.		

Table 2: Planning application executive summary.
Source: Spelthorne Borough Council (2025, p. 1).



Geovisualisation of Longcross vectors. Data sourced from OSM_AUDIT_2025 repository.

Synthesis: The Physical Body of the Cloud

An Eco-Marxist Critique of the Digital Spatial Fix

The Marxist Metabolic Rift: The forensic audit proves that “immaterial” cloud rendering requires the violent eradication of local biomass (NDVI Spatiotemporal Collapse).

The Digital Spatial Fix: Capital does not eliminate emissions; it displaces them. The film industry has anchored its ecological crisis into 13.2 hectares of asphalt and 17 high-voltage nodes.

The Thermodynamic Price: The +5 °C structural LST escalation is the direct, physical manifestation of Algorithmic Metabolism.

Accumulation by Dispossession: A £6,150 ecological buyout fee for a 392M in economic value reveals the true political economy of greenwashing.

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